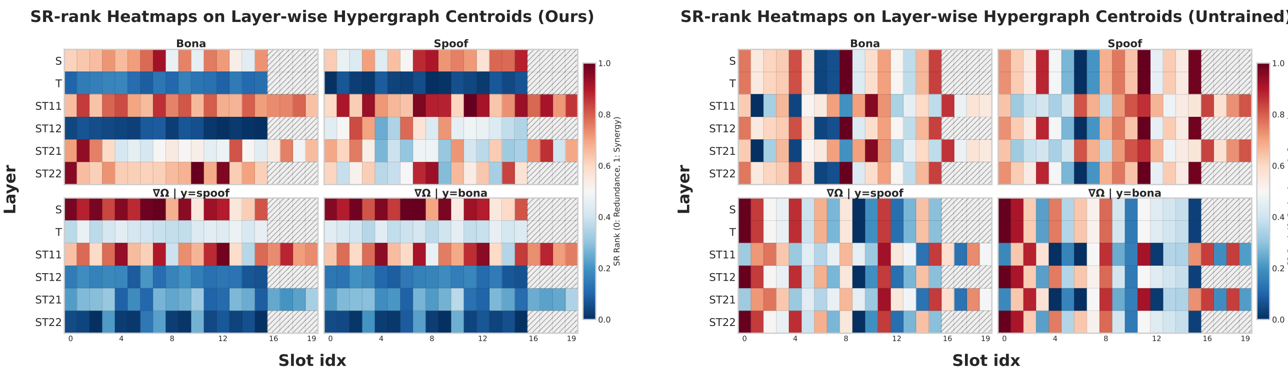
Table 9. Comparison of EERs (%) for different models trained on ASVspoof2019 LA. Lower is better. **Bold**, underlined, and † indicate the best, second-best, and third-best results, respectively. Subscripts denote the performance gaps between HyperPotter and its baseline.

Model	Params (M)	In-the-Wild	ASV20 19LA	ASV20 21LA	ASV20 21DF	ASV spoof5	FoR	Codecfake	ADD22 Track1	ADD22 Track3	ADD23 R1	ADD23 R2	Libri Voc	SONAR
Wav2Vec2+AASIST (Tak et al., 2022b)	317.84	11.19	0.221	0.82	6.63	16.24	7.46	43.36	31.04	16.52	27.74	21.92	11.13	46.12
XLSR+Conformer (Rosello et al., 2023)	302.40	-	-	0.97	2.58	-	-	-	-	-	-	-	-	-
XLSR+Conformer+TCM (Truong et al., 2024)	319	7.79	0.19†	3.00	2.15	18.85	10.69	36.01	37.40	20.94	23.43	22.74	2.35	26.57†
XLSR+AASIST2 (Zhang et al., 2024b)	-	-	0.15	1.61	2.77	-	-	-	-	-	-	-	-	-
WavLM+MFA (Guo et al., 2024)	-	-	0.42	5.08	2.56	-	-	-	-	-	-	-	-	-
XLSR+SLS (Zhang et al., 2024a)	340	7.45†	0.231	2.86	1.91	18.76	5.07†	**33.43**	33.95	15.74	**19.37**	21.09	**1.72**	24.72
XLSR+MoE (Wang et al., 2025b)	-	-	0.74	2.96	2.54	-	-	-	-	-	-	-	-	-
XLS-R+STCA+LMDC (Hao et al., 2025)	-	-	**0.09**	**0.78**	1.87	-	-	-	-	-	-	-	-	-
XLSR+Mamba (Xiao & Das, 2025)	319	6.70	0.421	0.93†	1.88†	14.40†	6.71	35.26†	34.22	19.36	21.84	**20.15**	2.15†	**24.26**
XLSR+BiCrossMamba (El Kheir et al., 2025)	318.21	7.94	0.71	3.83	2.35	13.67	6.85	37.70	**30.44**	18.69	29.44	29.93	2.12	27.36
Wav2Vec2+VIB(w/ HAR) (Doan et al., 2024)	-	3.78	2.68	-	2.07	-	10.32	-	-	-	-	-	-	-
Wav2Vec2+VIB(w/o HAR) (Doan et al., 2024)	-	9.78	0.57	-	3.38	-	-	-	-	-	-	-	-	-
Wav2Vec2+AASIST(Baseline)	317.84	7.58	0.26	2.48	4.08	**13.38**	4.24	40.22	33.79	16.14	25.55	22.21	6.96	32.02
HyperPotter	317.87	**5.72** $_{\downarrow 1.86}$	0.23$_{\downarrow 0.03}$	3.94$_{\uparrow 1.46}$	**1.78**$_{\downarrow 2.30}$	16.04$_{\uparrow 2.66}$	**3.89**$_{\downarrow 0.35}$	34.47$_{\downarrow 5.75}$	32.34† $_{\downarrow 1.45}$	**11.31**$_{\downarrow 4.83}$	21.49$_{\downarrow 4.06}$	21.84† $_{\downarrow 0.37}$	2.55$_{\downarrow 4.41}$	27.71$_{\downarrow 4.31}$

Note: HAR module of Wav2Vec2+VIB relies on re-synthesized augmentation (i.e., extra generated training data); therefore, we treat the w/o HAR variant as the primary fair comparison.



(a) Ours.

(b) Untrained.

Figure 9. Synergy-Redundancy rank heatmaps for global-prototype-aligned centroid slots across hypergraph layers: O-information (Ω) for spoof/bona-fide data and Gradient O-information conditioned on y ($\nabla\Omega \mid y; y \in \{\text{spoof}, \text{bona fide}\}$), with SR min–max normalized to $[0, 1]$ (per panel). High (red) values indicate that the centroid's interactions are predominantly synergistic, while low (blue) values indicate they are primarily redundant.

Table 10. Ablation study on HyperPotter sub-modules and alignment algorithms, reported as EERs (%).

Model	In-the-Wild	ASV20 19LA	ASV20 21LA	ASV20 21DF	ASV spoof5	FoR	Codecfake	ADD22 Track1	ADD22 Track3	ADD23 R1	ADD23 R2	Libri Voc	SONAR
Wav2Vec2+AASIST(Baseline)	7.58	0.26	2.48	4.08	**13.38**	4.24	40.22	33.79	16.14	25.55	22.21	6.96	32.02
HyperPotter	5.72	0.23	3.94	1.78	16.04	3.89	34.47	32.34	11.31	21.49	21.84	2.55	27.71
w/o Amplification Module (M2)	6.90 ↑	0.23	3.50 ↓	1.88 ↑	14.46 ↓	6.45 ↑	36.45 ↑	33.52 ↑	15.67 ↑	21.03 ↓	22.36 ↑	3.43 ↑	32.98 ↑
w/o A transposition (w/ A)	5.25 ↓	0.33 ↑	3.66 ↓	2.28 ↑	16.78 ↑	4.24 ↑	37.47 ↑	33.27 ↑	13.82 ↑	23.85 ↑	24.36 ↑	3.43 ↑	27.36 ↓
w/o $A^\top$ row-softmax (w/ col)	6.09 ↑	0.20 ↓	3.83 ↓	1.73 ↓	18.19 ↑	3.62 ↓	35.32 ↑	34.63 ↑	14.53 ↑	23.04 ↑	25.68 ↑	3.15 ↑	34.17 ↑
w/o Prototype Module (M3)	6.49 ↑	0.26 ↑	2.80 ↓	1.88 ↑	15.42 ↓	4.59 ↑	36.73 ↑	32.87 ↑	11.59 ↑	22.10 ↑	23.87 ↑	3.86 ↑	29.15 ↑
w/o Both Alignment Algs	5.95 ↑	0.19 ↓	3.83 ↓	2.25 ↑	19.07 ↑	4.46 ↑	36.41 ↑	34.39 ↑	13.19 ↑	23.60 ↑	23.39 ↑	2.86 ↑	30.12 ↑
w/o Soft Alignment Alg	7.45 ↑	0.23	3.25 ↓	2.05 ↑	19.42 ↑	4.02 ↑	37.02 ↑	33.28 ↑	14.03 ↑	24.35 ↑	28.64 ↑	3.81 ↑	30.24 ↑
w/o Slot Alignment Alg	6.18 ↑	0.14 ↓	3.24 ↓	2.07 ↑	19.90 ↑	4.02 ↑	35.11 ↑	35.32 ↑	13.95 ↑	22.31 ↑	24.10 ↑	2.61 ↑	31.00 ↑

Table 11. Development-set replacement experiments, where ITW is replaced by PartialSpoof v1 or ASVspoof5 (2k randomly selected samples), reported in EER (%).

Model	In-the-Wild	ASV20 19LA	ASV20 21LA	ASV20 21DF	ASV spoof5	FoR	Codecfake	ADD22 Track1	ADD22 Track3	ADD23 R1	ADD23 R2	Libri Voc	SONAR
Wav2Vec2+AASIST (ep80 | ITW dev)	7.58	0.26	2.48	4.08	13.38	4.24	40.22	33.79	16.14	25.55	22.21	6.96	32.02
HyperPotter (ep45 | ITW dev)	**5.72** $_{\downarrow 1.86}$	0.23$_{\downarrow 0.03}$	3.94$_{\uparrow 1.46}$	**1.78**$_{\downarrow 2.30}$	16.04$_{\uparrow 2.66}$	**3.89**$_{\downarrow 0.35}$	34.47$_{\downarrow 5.75}$	32.34† $_{\downarrow 1.45}$	**11.31**$_{\downarrow 4.83}$	21.49$_{\downarrow 4.06}$	21.84† $_{\downarrow 0.37}$	2.55$_{\downarrow 4.41}$	27.71$_{\downarrow 4.31}$
Wav2Vec2+AASIST (ep52 | PS dev)	8.79	0.27	2.25	4.62	12.94	4.68	40.93	34.60	15.07	25.02	21.13	8.58	33.03
HyperPotter (ep76 | PS dev)	6.67 ↓	0.31 ↑	3.58 ↑	2.00 ↓	17.38 ↑	4.64 ↓	36.32 ↓	33.70 ↓	11.97 ↓	20.85 ↓	20.66 ↓	3.82 ↓	26.04 ↓
Wav2Vec2+AASIST (ep52 | ASV5 dev)	9.26	0.34	2.79	4.48	12.77	5.96	41.12	35.26	15.45	24.84	21.67	9.90	36.75
HyperPotter (ep49 | ASV5 dev)	7.12 ↓	0.31 ↓	2.69↓	2.60 ↓	14.26 ↑	3.80 ↓	37.87 ↓	31.96 ↓	11.30 ↓	24.17 ↓	24.45 ↑	4.66 ↓	27.89 ↓

Table 12. Scalability experiments with longer audio inputs (8 s), reported in EER (%).

Model	**In-the-Wild**	**ASV20 19LA**	**ASV20 21LA**	**ASV20 21DF**	**ASV spoof5**	**FoR**	**Codecfake**	**ADD22 Track1**	**ADD22 Track3**	**ADD23 R1**	**ADD23 R2**	**Libri Voc**	**SONAR**
Wav2Vec2+AASIST (4s | ASV19)	7.58	0.26	2.48	4.08	13.38	4.24	40.22	33.79	16.14	25.55	22.21	6.96	32.02
HyperPotter (4s | ASV19)	**5.72** ↓	0.23↓	3.94↑	**1.78**↓	16.04↑	**3.89**↓	<u>34.47</u>↓	32.34†↓	**11.31**↓	<u>21.49</u>↓	21.84†↓	2.55↓	27.71↓
Wav2Vec2+AASIST (8s | ASV19)	8.22	0.41	4.10	2.66	19.52	4.55	38.61	37.30	11.94	17.93	16.54	5.32	39.72
HyperPotter (8s | ASV19)	5.81 ↓	0.17 ↓	2.04 ↓	2.15 ↓	14.63 ↓	3.40 ↓	35.89 ↓	32.45 ↓	12.76 ↑	20.10 ↑	21.16 ↑	2.81 ↓	35.50 ↓

Table 13. Evaluation on additional diverse datasets, reported in EER (%) and Accuracy (%). Accuracy is additionally reported for the fake-only dataset. All models are trained with ASVspoof 2019 LA.

Model	**Wav2Vec2+AASIST (Baseline)**	**HyperPotter**	**Wav2Vec2+VIB** (Doan et al., 2024)
Emofake *EER (%)*	2.01	**1.4**	-
MLAAD(+MAILABS) *EER (%)*	23.16	**18.23**	-
SingFake *EER (%)*	45.83	**40.77**	-
MLAAD *ACC (%)*	76.47	**81.68**	72.44

Table 14. Inference and training cost comparison between the Wav2Vec2-AASIST(Baseline) and HyperPotter on a server equipped with an AMD EPYC 7543 32-Core Processor and a single NVIDIA RTX A6000 GPU.

Batch size	Model	Inference			Training		
		Latency (ms) ↓	Throughput ↑	VRAM (MB) ↓	Iter. time (ms) ↓	Samples/s ↑	Iter./s ↑
1	Baseline	19.49	51.35	2630.64	101.33	9.87	9.87
	Ours	77.50	12.90	2631.64	264.31	3.78	3.78
8	Baseline	85.97	93.05	4718.64	289.67	27.62	3.45
	Ours	348.67	22.94	4720.64	1199.85	6.67	0.83
16	Baseline	162.23	98.36	6421.08	511.28	31.29	1.96
	Ours	661.64	24.12	6422.08	2240.85	7.14	0.45
64	Baseline	606.74	105.20	12922.32	1768.53	36.09	0.57
	Ours	2523.79	25.29	13352.32	8563.60	7.45	0.12

Table 15. Computational-cost analysis of HyperPotter under three controlled sweeps, conducted under the same hardware setup as Table 14. **Default configuration**: cluster setting $(16, 16, 20, 16, 20, 16)$, 4 s input, and `max_iter` $= 5$. **The default setting in each sweep is highlighted in bold for easy comparison.** Latency is reported as per-sample latency with batch size $= 1$; throughput and peak memory are reported with batch size $= 128$. For the cluster sweep, the six values in each configuration correspond to $(S, T, \mathrm{ST11}, \mathrm{ST12}, \mathrm{ST21}, \mathrm{ST22})$.

Experiment	Setting	Latency (ms) ↓	Throughput (utt/s) ↑	Peak Mem. (GiB) ↓	Mean Iter.	Mean Nodes
	Cluster-count sweep (4 s input, `max_iter` $= 5$)					
K	$0.25N$ $(11, 12, 14, 7, 14, 7)$	60.27	38.52	13.76	5.0	42.8
K	$0.50N$ $(21, 33, 27, 13, 27, 13)$	76.98	23.84	12.20	5.0	42.8
K	$0.75N$ $(32, 50, 41, 20, 41, 20)$	94.65	16.67	13.41	5.0	42.8
K	**HyperPotter** $(16, 16, 20, 16, 20, 16)$	**73.06**	**25.81**	**14.63**	**5.0**	**42.8**
	Node-count sweep (default cluster setting, `max_iter` $= 5$)					
N	1 s input	61.62	34.40	4.36	4.10	23.0
N	2 s input	62.03	34.99	7.49	5.00	29.5
N	**4 s input**	**69.34**	**27.20**	**14.97**	**5.00**	**42.8**
N	6 s input	74.85	23.75	22.46	5.00	55.7
N	8 s input	81.37	20.97	29.94	5.00	69.5
	FCM iteration sweep (4 s input, default cluster setting)					
T	max_iter = 1	58.51	27.48	13.76	1.0	42.8
T	max_iter = 3	66.73	27.43	12.20	3.0	42.8
T	**max_iter = 5**	**69.30**	**27.34**	**13.41**	**5.0**	**42.8**
T	max_iter = 7	76.53	27.16	14.63	7.0	42.8
T	max_iter = 10	86.32	26.79	15.85	10.0	42.8